TECHNICAL DILIGENCE REVIEW - RECORD 010

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-010

1. Terms of reference, method and interview programme

TDR-2025-34 is one of a series of technical reports; this one takes the reliability record, incident history and post-incident follow-through of the fleet platform and reports against the evidence available by 2026-01-09. Scope of the report at TDR-2025-34 is the reliability record, incident history and post-incident follow-through of the fleet platform, and the cut-off for evidence is 2026-01-09. 5 items in the report at TDR-2025-34 sit in a declined state, and the reviewer regards one of them as wrongly declined. Management's declinatures are reported in the present report at face value; the reviewer's disagreement is stated separately and does not overwrite them. The report at TDR-2025-34 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

The working method behind TDR-2025-34 was to start from the incident record and work back to the design, rather than the other way round. In the report at TDR-2025-34 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. TDR-2025-34 covers 20 of 34 services; the untested remainder is listed so a reader can see what was not reached. Sampling in the present report covered 20 of the 34 catalogued services and 6 of the 18 live sites. The reliability record, incident history and post-incident follow-through of the fleet platform produced more evidence than the report at TDR-2025-34 reproduces, and the working papers hold the remainder.

Calder Rowe Engineering Advisory asked the same four questions of every team interviewed for the report at TDR-2025-34, which is how the inconsistencies below were found. The interview programme behind this report covered 8 teams and produced 13 follow-up requests, of which most were satisfied. The present report did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. The approach R. Ostrowski took in TDR-2025-34 was to test the company's own assertions rather than to restate them. R. Ostrowski has kept the working papers behind the present report in a form that a second reviewer could re-perform.

The evidence convention in this report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. The convention throughout the present report is that an absence of evidence is reported as such rather than as evidence of absence. The report at TDR-2025-34 added two components to its sample after the first week, because the interview evidence pointed at them. For TDR-2025-34 the reviewer sampled 6 sites, including the largest deployment and the newest. Calder Rowe Engineering Advisory raised each finding with the owning engineer before it was written up, and the responses are recorded.

Two document requests remained partly unsatisfied when the report at TDR-2025-34 was delivered, and the affected conclusions are marked. R. Ostrowski did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. R. Ostrowski treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and the present report keeps the two apart. This report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. The evidence behind this report is retained by Calder Rowe Engineering Advisory and is available to Ardent Peak on request.

20 services were examined in detail for TDR-2025-34, chosen for their position in the dependency map rather than at random. R. Ostrowski chose the sample for the present report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. R. Ostrowski could not verify one class of claim in the present report, namely the behaviour of components deployed only at the two sites not visited. Where report TDR-2025-34 reports a supplier's test result, the reviewer read the certificate and did not witness the test. The reliability record, incident history and post-incident follow-through of the fleet platform is reported here at the level of detail Ardent Peak asked for and no deeper.

Schedule I - Interview programme and follow-up requests arising (TDR-2025-34)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-12-11	Marcus Duplessis	quality lead	Fleet Allocator	1	1
2026-06-23	Bettina Vogel	release manager	Site Provisioning Service	4	1
2026-07-13	Anjali Sekhon	release manager	Hardware-in-the-	5	5

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
			Loop Rig		
2026-03-16	Tomas Villareal	team lead	Video Snippet Service	2	2
2025-11-05	Sam Oyelaran	site reliability lead	Traffic Manager	1	1
2026-07-17	Yuki Harada	team lead	Identity and Authorization	1	0
2026-06-24	Anjali Sekhon	principal engineer	Charge Scheduler	4	3
2026-01-14	Priya Raghunathan	engineering owner	OTA Updater	5	2
2025-08-04	Anjali Sekhon	release manager	Alarm Router	5	4
2026-07-09	Tomas Villareal	head of engineering	Robot Shadow Store	3	1
2025-08-02	Grace Lindqvist	quality lead	Alarm Router	4	2
2026-10-03	Wanda Njoku	safety engineer	Analytics Warehouse Loader	2	2
2025-12-19	Wanda Njoku	quality lead	Fleet Health Monitor	1	0
2026-10-31	Colin Achebe	head of engineering	Video Snippet Service	1	1
2025-08-20	Lucia Ferrante	site reliability lead	Costmap Fusion	1	1
2026-07-04	Sonia Krall	security engineer	OTA Updater	4	2
2026-01-04	Colin Achebe	team lead	E-Stop Chain	4	2
2025-08-28	Priya Raghunathan	security engineer	Identity and Authorization	2	2
2026-07-08	Marcus Duplessis	field reliability lead	Lidar Driver	3	0
2026-06-15	Anjali Sekhon	principal engineer	Lidar Driver	1	0
2026-05-13	Wanda Njoku	field reliability lead	Fleet Health Monitor	4	0
2026-06-27	Marta Kellerman	engineering owner	E-Stop Chain	5	0
2025-10-16	Farida Cisse	team lead	Map Authoring Studio	2	2
2026-04-29	Nadia Semenova	site reliability lead	Simulation Harness	1	0
2026-05-14	Bettina Vogel	head of engineering	Map Authoring Studio	4	3
2025-08-13	Sam Oyelaran	security engineer	WMS Connector	1	1
2026-02-23	Ines Duarte	head of engineering	Video Snippet Service	5	4
2026-09-25	Peter Halvorsen	field reliability lead	Robot Shadow Store	5	3
2026-05-26	Wanda Njoku	principal engineer	E-Stop Chain	5	3
2026-06-29	Priya Raghunathan	security engineer	Speed and Separation Monitor	2	2

2. INC-2025-058 and the incident population examined

Grading incident INC-2025-058 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. WMS Connector incident INC-2025-058 records that on 2025-05-09, two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Incident INC-2025-058 was caused by a design gap: duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. Drawing the evidence in incident INC-2025-058 together, the same class of defect can recur in two sibling components that were not changed. The detection path for data integrity incident INC-2025-058 was the customer reported missing picks the same afternoon. incident INC-2025-058 is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

WMS Connector incident INC-2025-058 produced one commitment: to key suppression on the tenant and identifier pair and backfill the affected order records, owned by Tomas Villareal, dated 2025-07-15. Data integrity incident INC-2025-058 affected 2 site or tenant with no robot taken out of service. INC-2025-058 was mitigated in 2 h 35 m; the resolution ran to 21 days. INC-2025-058 is recorded as follows on delivery: landed on 2025-05-30; the backfill was completed on 2025-06-03. In taxonomy terms data integrity incident INC-2025-058 is a data integrity incident in WMS Connector. data integrity incident INC-2025-058 was checked against the corresponding entry in the company's own register, and the register was current.

How incident INC-2025-124 surfaced matters to the assessment: the console latency alert fired, though the underlying cause took a day to attribute. The trigger recorded for incident INC-2025-124 on 2025-10-01 was that a shard rebalance served shadow state up to ninety seconds stale without marking it stale. Landed on 2025-10-30; the load test was added on 2025-11-12 is the position on the INC-2025-124 commitment as at the review date. Grading consistency incident INC-2025-124 on evidence rather than on consequence gives limited, being a single account corroborated only by the ticket timeline. The commitment register shows incident INC-2025-124 against an undertaking to annotate

staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman named and 2025-11-30 committed. incident INC-2025-124 would be a candidate for the first post-close engineering review under any reasonable plan.

Contributing to multiple sites incident INC-2025-124, and not repaired by the remediation, is that no test exercised a shard rebalance while the cluster was under load. The judgement recorded against INC-2025-124 is that the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. 1 h 08 m elapsed before multiple sites incident INC-2025-124 was mitigated, with resolution at 29 days. Multiple sites felt multiple sites incident INC-2025-124, which affected 5 sites and 178 robots. Consistency is the class of Robot Shadow Store incident INC-2025-124; Robot Shadow Store is the component. INC-2025-124 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Following observability event INC-2026-028 the team undertook to group on the signal and block closure while a constituent alert remains active, with Wanda Njoku accountable by 2026-04-30. Strong, since the reviewer reproduced the failure mode in a laboratory environment is the evidence grade for Fleet Health Monitor incident INC-2026-028, and the reader should not read severity into it. Observability event INC-2026-028 took not a service outage; 2 days of delayed repair to stabilise and 31 days to close out. Observability event INC-2026-028 closure status is landed on 2026-04-10. Granite Peak Distribution incident INC-2026-028 reached the engineering team when the customer asked why one symptom persisted after the case had been closed. Granite Peak Distribution incident INC-2026-028 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Beyond the direct cause, INC-2026-028 shows that case closure did not check for unresolved constituent alerts. Scope of impact in INC-2026-028 was one site and 22 robots, taken from the fleet records rather than from the narrative. Taken with the rest of the record, INC-2026-028 indicates that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. In taxonomy terms observability incident INC-2026-028 is a observability incident in Fleet Health Monitor. What went wrong in INC-2026-028 was that alert grouping keyed on site and severity rather than on the underlying signal. observability incident INC-2026-028 was excluded from an earlier draft of TDR-2025-34 and reinstated after the reviewer obtained the supporting record.

On 2026-04-21 the record for incident INC-2026-045 opens: a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. After Evergreen Warehousing incident INC-2026-045, Sam Oyelaran took the action to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current hardware by 2026-06-30. As to whether the Edge Gateway Agent incident INC-2026-045 remediation actually landed: the alert landed on 2026-05-17; the buffer extension is in progress. Evergreen Warehousing incident INC-2026-045 was detected because the buffer pressure metric was visible per robot but no site-level alert existed. A contributing factor in incident INC-2026-045 was that buffer pressure was never aggregated to a site view, so the condition was not observable. Evergreen Warehousing incident INC-2026-045 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

INC-2026-045 is not read as an isolated failure: the remediation addresses the immediate cause and leaves the contributing factor in place. Mitigation of incident INC-2026-045 took 5 h 12 m and full resolution took 26 days. Evergreen Warehousing felt INC-2026-045, which affected one site and 39 robots. Evergreen Warehousing incident INC-2026-045 carries strong on the cause and limited on the contributing factor evidence, and the reviewer declined to raise the finding beyond what that supports. Cause analysis in incident INC-2026-045 concluded that the buffer is sized for four hours and drops oldest-first without raising an alert. Evergreen Warehousing incident INC-2026-045 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Verification of the tooling event INC-2026-021 remediation gives not landed; the determinism check is written but not enabled. On 2026-02-24 the record for internal incident INC-2026-021 opens: a nightly simulation suite produced non-reproducible results after the physics core was rebuilt. Contributing to internal incident INC-2026-021, and not repaired by the remediation, is that the build pinned the compiler but not the optimisation flags used for release builds. Incident INC-2026-021 produced one commitment: to pin the full build configuration and add a determinism check to the harness, owned by Colin Achebe, dated 2026-05-31. Not applicable; developer tooling elapsed before incident INC-2026-021 was mitigated, with resolution at open at the review date. tooling incident INC-2026-021 rests on a single source, and the reviewer has said so rather than implying corroboration.

In taxonomy terms Simulation Harness incident INC-2026-021 is a tooling incident in Simulation Harness. The affected population in tooling incident INC-2026-021 was no production site and no robot directly, the exposure being to internal systems. The reviewer grades the internal incident INC-2026-021 evidence as strong, resting on the production traces the team supplied and has not upgraded the finding to compensate. Internal incident INC-2026-021 traces back to a single fact: the physics core is not deterministic across optimisation levels. Tooling incident INC-2026-021 reached the engineering team when the autonomy team noticed a step change in the pass rate. tooling event INC-2026-021 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

INC-2026-009 is not read as an isolated failure: the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Calibration event INC-2026-009 touched one site and 4 robots, and Blue Mesa Components was the affected account. Blue Mesa Components incident INC-2026-009 would have been shorter or smaller had it not been that the service workflow allowed the step to be marked complete without a calibration record. Cause analysis in calibration event INC-2026-009 concluded that force limiting depends on a calibration that is only triggered by a service workflow step. Calibration event INC-2026-009 shows a short mitigation and a long tail: 1 h 30 m to quarantine the affected robots against 48 days. calibration incident INC-2026-009 is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

The detection path for Gripper Controller incident INC-2026-009 was the payload damage was reported by the customer. 2026-01-27 is the date of Gripper Controller incident INC-2026-009, on which a gripper applied more force than the configured limit after a mechanical replacement without recalibration. Grading calibration incident INC-2026-009 on evidence rather than on consequence gives strong, resting on the production traces the team supplied. Ines Duarte committed after Blue Mesa Components incident INC-2026-009 to require a calibration record before returning a manipulator robot to service, due 2026-04-15. For INC-2026-009 the follow-through was tested rather than assumed, and landed on 2026-03-16 with a workflow gate. calibration incident INC-2026-009 is one of the entries where TDR-2025-34 relies on the company's own measurement rather than an independent one.

The post-mortem for INC-2025-047 identifies the cause as follows: the driver did not re-derive the sensor time offset on reconnect. The driver fix landed on 2025-04-05; the second rig was still unmodified at review is the position on the timing event INC-2025-047 commitment as at the review date. Timing incident INC-2025-047 rests on limited; the log retention window had closed before the reviewer asked for the raw data evidence, which the reviewer states separately from the severity of the finding. Lidar Driver incident INC-2025-047 would have been shorter or smaller had it not been that the bench rig that would have caught it carries a different lidar variant. The corrective action for Asterion Logistics incident INC-2025-047 is to re-derive the time offset on every reconnect and add the affected variant to the second bench rig, and 2025-06-15 is the date the team gave. incident INC-2025-047 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Asterion Logistics felt Asterion Logistics incident INC-2025-047, which affected one site and 12 robots. In incident INC-2025-047 the first signal was an automated fusion timing alert fired inside four minutes, and the reviewer treats that as alerted detection. Assessed on its own record, Lidar Driver incident INC-2025-047 shows that the fix is real and the regression protection is not, so recurrence remains possible. Lidar Driver incident INC-2025-047 was mitigated in 38 m; the resolution ran to 3 days. Incident INC-2025-047 was raised on 2025-04-02 after a lidar unit reconnected after a cable seating fault and reported frames with a stale timestamp offset. timing event INC-2025-047 is reported at the evidence strength available and has not been promoted to fit the narrative.

Attention came to internal incident INC-2026-062 when the release audit found it during the quarterly review. So far as reach is concerned, process control incident INC-2026-062 involved no production site and no robot directly, the exposure being to internal systems. On delivery, process control incident INC-2026-062 shows this: not landed; the change is scoped but not scheduled. On Release Orchestrator incident INC-2026-062 the evidence is moderate; the narrative is consistent but the reviewer could not independently reproduce it; the severity assessment is stated separately below. What process control event INC-2026-062 tells the reviewer is that the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. process control incident INC-2026-062 is reported at the evidence strength available and has not been promoted to fit the narrative.

The remediation committed after process control incident INC-2026-062 was to require a justification and a second approver within one working day for every emergency release, owned by Colin Achebe with a due date of 2026-08-31.

Incident INC-2026-062 classifies as a process control event against Release Orchestrator. Not applicable; process only elapsed before process control event INC-2026-062 was mitigated, with resolution at open at the review date. The engineering team's root cause for process control event INC-2026-062 is that the emergency path is available to any release engineer without a justification field, and the reviewer found the analysis sound. In process control incident INC-2026-062, the policy defines an emergency but the tool does not ask which condition applies, which is a process finding rather than a code finding. incident INC-2026-062 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Schedule F - Incident population examined, with follow-through status (TDR-2025-34)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorizatio n	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

2. INC-2025-058 and the incident population examined - part 2 of 2

This is part 2 of the 2. INC-2025-058 and the incident population examined, carried over unbroken; a reader reconciling a reference should treat the parts as a single register. No row is omitted at the join.

Definitions given once for the 2. INC-2025-058 and the incident population examined are relied on again rather than restated, the compilation basis being identical. No row is omitted at the join.

Schedule F - Incident population examined, with follow-through status (TDR-2025-34) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-139	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed
INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable;	11 days	process defect	landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
					caught before billing			
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

3. Detection paths and time to mitigate

Remediation for OTA Updater incident INC-2025-084 is to persist the pause state and require a design note for any change to the release state machine; Erik Palmgren owns it and the committed date is 2025-08-15. Scope of impact in Blue Mesa Components incident INC-2025-084 was one site and 11 robots, taken from the fleet records rather than from the narrative. As to whether the state persistence event INC-2025-084 remediation actually landed: landed on 2025-07-14. The post-mortem for OTA Updater incident INC-2025-084 identifies the cause as follows: the pause flag was held in process memory rather than in the release state record. OTA Updater owns OTA Updater incident INC-2025-084, which the reviewer classes as state persistence. Blue Mesa Components incident INC-2025-084 was one of two items where TDR-2025-34 could not obtain the underlying data within the engagement.

The reviewer grades the Blue Mesa Components incident INC-2025-084 evidence as moderate, resting on the team's own post-mortem and one supporting log extract and has not upgraded the finding to compensate. Beyond the direct cause, OTA Updater incident INC-2025-084 shows that the pause control had been added in a hotfix without a design review. The detection path for state persistence event INC-2025-084 was the site team escalated when robots

rebooted outside the maintenance window. Time to mitigate state persistence event INC-2025-084 was 27 m, and time to resolve was 5 days. State persistence incident INC-2025-084 supports one conclusion and not another: the remediation addresses the immediate cause and leaves the contributing factor in place. OTA Updater incident INC-2025-084 was reviewed against the position twelve months earlier, and nothing measurable has changed.

On secret exposure event INC-2025-118 the evidence is strong on the cause and limited on the contributing factor; the severity assessment is stated separately below. The corrective action for secret exposure incident INC-2025-118 is to redact authorization headers at the logging layer and cut build log retention to fourteen days, and 2025-10-31 is the date the team gave. Secret exposure event INC-2025-118 is recorded as follows on delivery: redaction landed on 2025-09-19; retention was changed on 2025-09-24. Identity and Authorization incident INC-2025-118 would have been shorter or smaller had it not been that build logs were retained far longer than the token lifetime, widening the exposure window. Detection came through the dependency and secret scanner flagged it on the following scan, which places secret exposure incident INC-2025-118 on the alerted side of the ledger. secret exposure event INC-2025-118 was checked against the company's own register, which had not been updated since the change landed.

Incident INC-2025-118 belongs to the secret exposure group and sits with Identity and Authorization. Internal felt secret exposure event INC-2025-118, which affected no production site and no robot directly, the exposure being to internal systems. Secret exposure incident INC-2025-118 was mitigated in 3 h 40 m to revoke and rotate; the resolution ran to 7 days. INC-2025-118 began on 2025-09-17 when a service token with a twelve-hour lifetime was found in a build log retained for ninety days. Underneath INC-2025-118 lies the design decision that a debug logging statement in the connector build printed the full authorization header. internal incident INC-2025-118 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Ironwood Grocery Group incident INC-2025-103 was detected because the queue depth alert introduced after the March incident fired correctly. So far as reach is concerned, Ironwood Grocery Group incident INC-2025-103 involved one site and 2 robots. The remediation committed after Ironwood Grocery Group incident INC-2025-103 was to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, owned by Priya Raghunathan with a due date of 2025-10-31. The two durations recorded for Motion Planner incident INC-2025-103 are 19 m to mitigate and 58 days to resolve. Motion Planner accounts for INC-2025-103, a algorithm defect event. incident INC-2025-103 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Grading INC-2025-103 on evidence rather than on consequence gives strong, since the reviewer reproduced the failure mode in a laboratory environment. Root cause in Ironwood Grocery Group incident INC-2025-103 was that the yield heuristic had no tie-break and both robots computed the same deference score. Motion Planner incident INC-2025-103 carries a second-order finding: the simulation scenario for this geometry existed but had been excluded from the nightly suite. Assessed on its own record, algorithm defect incident INC-2025-103 shows that the same class of defect can recur in two sibling components that were not changed. On delivery, incident INC-2025-103 shows this: the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. Motion Planner incident INC-2025-103 was one of two items where TDR-2025-34 could not obtain the underlying data within the engagement.

As to whether the incident INC-2026-057 remediation actually landed: not landed; data collection is running and the imbalance alert is in review. Battery Management Service incident INC-2026-057 was contained to 2 sites and 5 robots. Following Dunbar Industrial incident INC-2026-057 the team undertook to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, with Peter Halvorsen accountable by 2026-09-30. Cause analysis in incident INC-2026-057 concluded that the estimator was fitted on packs that follow a full charge cycle. The detection path for incident INC-2026-057 was a robot derated mid-shift and the pack was pulled for analysis. modelling event INC-2026-057 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

In taxonomy terms incident INC-2026-057 is a modelling incident in Battery Management Service. Incident INC-2026-057 was raised on 2026-05-19 after state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Battery Management Service incident INC-2026-057 was mitigated in not a service outage; robots derated; the resolution ran to open at the review date. Modelling incident INC-2026-057 supports one conclusion and not another: the incident is unremarkable in itself and revealing about the observability of the surrounding system. Strong, resting on the production traces the team supplied is the evidence grade for Dunbar Industrial incident

INC-2026-057, and the reader should not read severity into it. incident INC-2026-057 was checked against the corresponding entry in the company's own register, and the register was current.

Process gap incident INC-2025-096 began on 2025-08-05 when an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. The remediation committed after INC-2025-096 was to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, owned by Wanda Njoku with a due date of 2025-10-15. Alarm Router incident INC-2025-096 affected no production site and no robot directly, the exposure being to internal systems. INC-2025-096 traces back to a single fact: the routing table fell back to a mailbox rather than to a secondary responder. The detection path for incident INC-2025-096 was discovered the next morning when the alarm was found unacknowledged. internal incident INC-2025-096 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The contributing factor recorded against incident INC-2025-096 is that the rotation source of truth was a spreadsheet maintained outside the alarm router. The reviewer grades the INC-2025-096 evidence as moderate; the narrative is consistent but the reviewer could not independently reproduce it and has not upgraded the finding to compensate. Alarm Router incident INC-2025-096 shows a short mitigation and a long tail: not applicable; no service impact against 26 days. INC-2025-096 belongs to the process gap group and sits with Alarm Router. What internal incident INC-2025-096 tells the reviewer is that management's severity rating is defensible and its remediation priority is not. process gap event INC-2025-096 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

As to whether the performance event INC-2026-004 remediation actually landed: not landed; the matcher replacement is in design and the re-survey is scheduled. The commitment register shows performance event INC-2026-004 against an undertaking to replace the fallback matcher and re-survey the affected aisles, with Rahul Menon named and 2026-06-30 committed. For INC-2026-004 the reviewer holds moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence and reports the finding at that strength. The blast radius of incident INC-2026-004 was one site and 18 robots. The reviewer treats one contributing factor in incident INC-2026-004 as the more important lesson: the site was surveyed with a sparse feature set that increases cache misses. incident INC-2026-004 was excluded from an earlier draft of TDR-2025-34 and reinstated after the reviewer obtained the supporting record.

Mitigation of Kestrel Medical Supply incident INC-2026-004 took not a service outage; 5 days to a workaround and full resolution took open at the review date. Kestrel Medical Supply incident INC-2026-004 was raised on 2026-01-13 after relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. Assessed on its own record, performance incident INC-2026-004 shows that the customer-reported detection path is the finding, not the defect itself. In taxonomy terms Kestrel Medical Supply incident INC-2026-004 is a performance incident in Localization Engine. What went wrong in performance incident INC-2026-004 was that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Localization Engine incident INC-2026-004 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-34.

Incident INC-2025-139 was caused by a design gap: control and telemetry traffic shared one topic and one consumer group budget. A contributing factor in Event Bus incident INC-2025-139 was that the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. In taxonomy terms Event Bus incident INC-2025-139 is a capacity incident in Event Bus. Event Bus incident INC-2025-139 shows a short mitigation and a long tail: 2 h 47 m against 63 days. For capacity event INC-2025-139 the follow-through was tested rather than assumed, and the split landed on 2026-01-04; the quarterly review has not yet run. capacity incident INC-2025-139 carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

Taken with the rest of the record, INC-2025-139 indicates that the blast radius was set by a deployment topology decision made years earlier. The blast radius of incident INC-2025-139 was 7 sites and 246 robots. For capacity event INC-2025-139 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. Incident INC-2025-139 records that on 2025-11-02, a telemetry backlog on a shared topic delayed control events by up to eleven seconds. Following multiple sites incident INC-2025-139 the team undertook to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly, with Erik Palmgren accountable by 2026-01-31. capacity event INC-2025-139 was checked against the company's own register, which had not been updated since the change landed.

Traffic Manager incident INC-2025-041 touched 2 sites and 61 robots, and Crestline Fulfillment was the affected account. In taxonomy terms resource leak incident INC-2025-041 is a resource leak incident in Traffic Manager. What incident INC-2025-041 tells the reviewer is that the remediation addresses the immediate cause and leaves the contributing factor in place. Incident INC-2025-041 took 1 h 52 m to stabilise and 9 days to close out. Following incident INC-2025-041 the team undertook to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, with Owen Brantley accountable by 2025-05-30. incident INC-2025-041 carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

Traffic Manager incident INC-2025-041 records that on 2025-03-14, a robot lost its network link mid-traverse and its corridor reservation was never released. Root cause in resource leak incident INC-2025-041 was that the reservation lease had no expiry, so release depended on the departing robot. Verification of the incident INC-2025-041 remediation gives landed on 2025-05-22, verified in the reservation replay suite. The reviewer treats one contributing factor in Traffic Manager incident INC-2025-041 as the more important lesson: the alert rule for queue depth existed but was scoped to a metric the site did not emit. For resource leak event INC-2025-041 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. incident INC-2025-041 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

4. Root causes, contributing factors and recurrence risk

Evidence strength on the incident INC-2025-078 analysis is strong, since the reviewer reproduced the failure mode in a laboratory environment. On 2025-06-24 the record for Harborline Systems incident INC-2025-078 opens: the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Harborline Systems incident INC-2025-078 was detected because an internal engineer noticed the discrepancy during a demonstration. Incident INC-2025-078 belongs to the stale state group and sits with Remote Operations Console. Mitigation of stale state event INC-2025-078 took 1 h 15 m and full resolution took 34 days. incident INC-2025-078 was excluded from an earlier draft of TDR-2025-34 and reinstated after the reviewer obtained the supporting record.

Remote Operations Console incident INC-2025-078 touched 4 site or tenant with no robot taken out of service, and Harborline Systems was the affected account. What incident INC-2025-078 tells the reviewer is that the fix is real and the regression protection is not, so recurrence remains possible. Verification of the stale state event INC-2025-078 remediation gives landed on 2025-07-28; the regression test was still marked flaky at review. Remediation for incident INC-2025-078 is to invalidate the roster cache on a fleet membership event and add a session-level regression test; Anjali Sekhon owns it and the committed date is 2025-08-31. The contributing factor recorded against stale state event INC-2025-078 is that no test covered a roster change during an open operator session. incident INC-2025-078 rests on a single source, and the reviewer has said so rather than implying corroboration.

Hardware fault incident INC-2025-090 closure status is the inspection change landed on 2025-08-04; nine units await retrofit. The classification applied to Fathom Mobility incident INC-2025-090 is hardware fault, and the affected component is Safety Controller. Reading incident INC-2025-090 from its first entry, on 2025-07-19 a protective stop was triggered by a false positive on one safety channel while the second channel read clear. Mitigation of Safety Controller incident INC-2025-090 took immediate stop, 22 m to return to service and full resolution took 41 days. The judgement recorded against hardware fault event INC-2025-090 is that the fix is real and the regression protection is not, so recurrence remains possible. Safety Controller incident INC-2025-090 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Incident INC-2025-090 was caused by a design gap: a marginal connector on one channel produced intermittent readings inside the accepted noise band. Limited; the log retention window had closed before the reviewer asked for the raw data is the evidence grade for hardware fault incident INC-2025-090, and the reader should not read severity into it. Fathom Mobility incident INC-2025-090 carries a second-order finding: the connector torque specification was not part of the incoming inspection sheet. The detection path for Fathom Mobility incident INC-2025-090 was the robot self-reported the fault and the safety case review picked it up. Bettina Vogel committed after hardware fault event INC-2025-090 to add the connector torque check to incoming inspection and retrofit the affected harness batch, due 2025-09-30. hardware fault event INC-2025-090 illustrates why TDR-2025-34 separates the severity of a finding from the confidence behind it.

Larkspur Home incident INC-2025-146 belongs to the process defect group and sits with Billing Usage Meter. Following Larkspur Home incident INC-2025-146 the team undertook to fold the reset into the re-run command and make double-counting detectable in the ledger, with Nadia Semenova accountable by 2025-12-31. As to whether the process defect event INC-2025-146 remediation actually landed: landed on 2025-11-30. The two durations recorded for process defect incident INC-2025-146 are not applicable; caught before billing to mitigate and 11 days to resolve. Process defect event INC-2025-146 was raised on 2025-11-19 after a re-run of a failed metering window double-counted usage for one customer. INC-2025-146 is reported here in full because a summary of it in an earlier draft of TDR-2025-34 was misread as an assurance.

The reviewer treats one contributing factor in INC-2025-146 as the more important lesson: the runbook described the reset in prose rather than making it part of the command. For process defect event INC-2025-146 the reviewer holds strong on the cause and limited on the contributing factor evidence and reports the finding at that strength. The engineering team's root cause for INC-2025-146 is that the re-run procedure required a manual ledger reset that the operator skipped, and the reviewer found the analysis sound. The affected population in INC-2025-146 was no production site and no robot directly, the exposure being to internal systems. Taken with the rest of the record, Billing Usage Meter incident INC-2025-146 indicates that the remediation addresses the immediate cause and leaves the contributing factor in place. Billing Usage Meter incident INC-2025-146 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

INC-2026-015 touched 2 sites and 7 robots, and Fathom Mobility was the affected account. Fathom Mobility incident INC-2026-015 carries moderate on the cause and strong on the delivery position, which was checked in the repository evidence, and the reviewer declined to raise the finding beyond what that supports. Duration for validation incident INC-2026-015 splits into 44 m of impact and 22 days to close the underlying cause. Config Distribution accounts for validation incident INC-2026-015, a validation event. The contributing factor recorded against Config Distribution incident INC-2026-015 is that the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed. INC-2026-015 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

2026-02-09 is the date of Config Distribution incident INC-2026-015, on which a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. Detection came through the on-robot supervisor raised a validation fault on the next mission, which places Config Distribution incident INC-2026-015 on the reported side of the ledger. The reviewer checked the validation event INC-2026-015 commitment against the code and release records; landed on 2026-03-03. Incident INC-2026-015 was caused by a design gap: precondition validation ran at publication time rather than at apply time. Remediation for INC-2026-015 is to move precondition validation to apply time and add an offline apply case to the platform suite; Sam Oyelaran owns it and the committed date is 2026-03-31. validation event INC-2026-015 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The reviewer grades the incident INC-2026-033 evidence as strong on the cause and limited on the contributing factor and has not upgraded the finding to compensate. The immediate event in INC-2026-033 was that a speed reduction engaged later than expected when an operator approached faster than the assumed walking speed, dated 2026-03-25. The reviewer's reading of safety margin incident INC-2026-033 is that the team's analysis was honest about a gap it could have concealed. The contributing factor recorded against INC-2026-033 is that the assumption was documented in the internal safety case and accepted at the time. What went wrong in Asterion Logistics incident INC-2026-033 was that the protective distance uses a fixed assumed approach speed rather than a measured closing rate. safety margin event INC-2026-033 is the entry that changed most between the first and second document requests behind TDR-2025-34.

Time to mitigate INC-2026-033 was immediate speed reduction; 3 h to review the fleet, and time to resolve was open at the review date. In taxonomy terms INC-2026-033 is a safety margin incident in Speed and Separation Monitor. Incident INC-2026-033 was detected because the on-robot safety log flagged a margin breach that did not become a contact event. So far as reach is concerned, incident INC-2026-033 involved one site and 6 robots. Verification of the Speed and Separation Monitor incident INC-2026-033 remediation gives not landed; a design study is running and the vendor assessment is booked. incident INC-2026-033 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The immediate event in incident INC-2026-039 was that an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour, dated 2026-04-07. Remediation for Crestline Fulfillment incident INC-2026-039 is to serialise cancellation against assignment and add an hourly orphan sweep; Grace Lindqvist owns it and the committed date is 2026-05-31. Drawing the evidence in incident INC-2026-039 together, the failure was in the detection design rather than in the code that failed. Evidence strength on the concurrency event INC-2026-039 analysis is limited, being a single account corroborated only by the ticket timeline. 1 h 04 m elapsed before incident INC-2026-039 was mitigated, with resolution at 19 days. Order Broker incident INC-2026-039 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

So far as reach is concerned, concurrency incident INC-2026-039 involved one site and 1 robots. Order Broker owns incident INC-2026-039, which the reviewer classes as concurrency. As to whether the Crestline Fulfillment incident INC-2026-039 remediation actually landed: landed on 2026-04-26. Order Broker incident INC-2026-039 traces back to a single fact: cancellation and assignment wrote to the task record without a shared lock. In concurrency incident INC-2026-039 the first signal was the site supervisor reported a blocked location, and the reviewer treats that as reported detection. incident INC-2026-039 was checked against the company's own register, which had not been updated since the change landed.

Schedule G - Open defect population and ageing by service (TDR-2025-34)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2025-09-13	0 high, 2 medium, 12 low	2	Priya Raghunathan
Costmap Fusion	21	2025-09-20	1 high, 1 medium, 19 low	11	Daniel Okafor
Fleet Allocator	9	2025-12-05	1 high, 1 medium, 7 low	10	Marta Kellerman
Traffic Manager	12	2025-05-17	0 high, 4 medium, 8 low	3	Owen Brantley
Charge Scheduler	7	2025-05-13	2 high, 1 medium, 4 low	11	Nadia Semenova
Safety Controller	3	2025-04-24	0 high, 1 medium, 2 low	7	Bettina Vogel
E-Stop Chain	2	2025-07-17	1 high, 1 medium, 0 low	12	Theo Bassett
Lidar Driver	18	2025-06-12	2 high, 7 medium, 9 low	1	Yuki Harada
Localization Engine	16	2026-03-31	1 high, 4 medium, 11 low	4	Rahul Menon
WMS Connector	23	2025-04-16	0 high, 2 medium, 21 low	14	Tomas Villareal
Order Broker	11	2025-05-09	0 high, 2 medium, 9 low	14	Grace Lindqvist
Telemetry Ingest	19	2025-12-17	2 high, 5 medium, 12 low	1	Farida Cisse
OTA Updater	8	2025-04-28	0 high, 1 medium, 7 low	6	Erik Palmgren
Identity and Authorization	6	2026-02-15	0 high, 2 medium, 4 low	3	Sonia Krall
Remote Operations Console	27	2025-03-02	2 high, 9 medium, 16 low	2	Anjali Sekhon
Simulation Harness	31	2025-06-29	0 high, 6 medium, 25 low	5	Colin Achebe
Hardware-in-the-Loop Rig	13	2026-04-02	1 high, 6 medium, 6 low	12	Nathan Oduya

4. Root causes, contributing factors and recurrence risk - part 2 of 2

Presentation of the 4. Root causes, contributing factors and recurrence risk carries on below without restarting the row numbers, the break being one of pagination only. The break is editorial.

Whatever caveat the 4. Root causes, contributing factors and recurrence risk carried at its head travels with these rows, and no additional caveat attaches to this part. No row is omitted at the join.

Schedule G - Open defect population and ageing by service (TDR-2025-34) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2026-04-13	2 high, 6 medium, 9 low	14	Wanda Njoku
Battery Management Service	10	2025-08-26	2 high, 5 medium, 3 low	1	Peter Halvorsen
Gripper Controller	15	2025-07-01	1 high, 2 medium, 12 low	9	Ines Duarte
Map Authoring Studio	22	2025-11-23	2 high, 8 medium, 12 low	12	Lucia Ferrante
Site Provisioning Service	5	2025-11-01	2 high, 1 medium, 2 low	3	Marcus Duplessis
Event Bus	4	2025-07-29	0 high, 1 medium, 3 low	7	Erik Palmgren
Config Distribution	6	2026-05-05	2 high, 1 medium, 3 low	4	Sam Oyelaran
Robot Shadow Store	9	2025-03-15	2 high, 2 medium, 5 low	2	Marta Kellerman
Analytics Warehouse Loader	20	2025-05-02	1 high, 8 medium, 11 low	10	Farida Cisse
Billing Usage Meter	7	2025-06-30	1 high, 2 medium, 4 low	6	Nadia Semenova
Alarm Router	14	2026-03-17	2 high, 2 medium, 10 low	14	Wanda Njoku
Video Snippet Service	12	2025-08-06	2 high, 6 medium, 4 low	13	Anjali Sekhon
Speed and Separation Monitor	4	2026-01-08	1 high, 2 medium, 1 low	12	Bettina Vogel
Wheel Odometry Filter	8	2026-01-23	1 high, 4 medium, 3 low	14	Yuki Harada
Payload Scale Interface	11	2026-04-25	1 high, 2 medium, 8 low	13	Ines Duarte
Edge Gateway Agent	5	2025-11-18	0 high, 2 medium, 3 low	2	Sam Oyelaran
Release Orchestrator	16	2025-04-25	1 high, 6 medium, 9 low	6	Colin Achebe

5. Reliability of the services carrying the incidents

Billing Usage Meter was assessed on the metering specification, three monthly reconciliations and the ledger immutability check, and nothing beyond that was available in the data room. Data and Telemetry publishes one objective for Billing Usage Meter: metered usage reconciles to the order journal within 0.5 per cent each month. Interface discipline around Billing Usage Meter is adequate; the schema is versioned and change is gated on a compatibility check. 7 items sit in the Billing Usage Meter backlog; the entry that bears on the investment case is that a re-run of a failed metering window can double-count if the operator skips the reset step. Billing Usage Meter's dependency ageing was checked: none identified. Billing Usage Meter was checked against the corresponding entry in the company's own register, and the register was current.

Billing Usage Meter was built in Go 1.22 and has not been rewritten; it still runs on cloud, containers on a managed Kubernetes cluster against PostgreSQL 14 with an immutable usage ledger. Billing Usage Meter sits at 79 per cent line coverage; the trend across the reviewed period was up by three points. Billing Usage Meter ships monthly, aligned to the billing cycle; the reviewer sampled the last twelve releases. The finance export and the customer usage report rely on Billing Usage Meter, which in turn relies on the order broker and the WMS connector. Billing Usage Meter is owned by the Data and Telemetry team, with Nadia Semenova named as the engineering owner of record. Billing Usage Meter was cross-checked against the engineering record supplied for TDR-2025-34, and the two agree.

Taking the evidence together, Lidar Driver the reviewer accepts the engineering position while noting that the verifying test does not exist. Automated test coverage on Lidar Driver stands at 44 per cent line coverage. Lidar Driver was built in C++17 and has not been rewritten; it still runs on on-robot Linux, user-space driver against no store; frames are handed to fusion in memory. Where Lidar Driver is concerned, the commitment is that no more than one dropped scan per ten thousand at nominal spin rate; the reviewer traced the metric to its source. Lidar Driver runs edge, one process per

sensor; that placement is what determines whether a fault is one robot or one region. Lidar Driver is the clearest instance in TDR-2025-34 of a control that exists on paper and is weaker in practice.

Lidar Driver was tested against the dependency ageing register, which records the vendor's software development kit is two major versions behind current. Lidar Driver carries 18 open defects, among them a case where reconnect handling leaves the timestamp offset uncorrected for the first three frames. Lidar Driver is owned by the Robot Firmware team, with Yuki Harada named as the engineering owner of record. What the reviewer had for Lidar Driver was the driver source, the vendor integration guide and a captured reconnect trace; the conclusions below do not reach past it. The costmap fusion pipeline and the localization engine rely on Lidar Driver, which in turn relies on the sensor hardware and the time synchronisation daemon. Lidar Driver is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Order Broker sits at 78 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. 11 items sit in the Order Broker backlog; the entry that bears on the investment case is that order cancellation races with assignment and can leave an orphaned task record. The cadence recorded for Order Broker is weekly, which matches the observed release history. Evidence examined for Order Broker comprised the broker journal schema, the nightly reconciliation output and two cancellation incidents. Upstream of Order Broker sit the WMS connector and the site configuration service; downstream, the fleet allocator and the analytics warehouse loader consume its output. Order Broker is treated as material to the investment case rather than as a routine engineering item.

Hosting for Order Broker is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with an event journal. The ageing position for Order Broker is none identified, and the reviewer checked it against the vendor statements on file. Order Broker does not share a schema with any consumer; every read by the fleet allocator and the analytics warehouse loader goes through the published interface. Fleet Services publishes one objective for Order Broker: no accepted order is lost, measured by journal reconciliation each night. Order Broker runs cloud, regional; that placement is what determines whether a fault is one robot or one region. Order Broker was checked against the corresponding entry in the company's own register, and the register was current.

14 items sit in the Alarm Router backlog; the entry that bears on the investment case is that an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. Alarm Router owns its own data and exposes it only through a versioned interface, which the reviewer read. Upstream of Alarm Router sit the fleet health monitor and the on-call rotation source; downstream, the paging provider and the customer notification channel consume its output. The reviewer's judgement on Alarm Router is that the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. For Alarm Router the stated target is that a dispatched alarm reaches an on-call responder inside 90 s at p95, expressed against production traffic rather than a benchmark. Alarm Router is one of the entries where TDR-2025-34 relies on the company's own measurement rather than an independent one.

For Alarm Router the reviewer worked from the routing table export, delivery receipts for one quarter and two missed-page reviews. Field Reliability releases Alarm Router fortnightly, and no release in the sampled window bypassed the gate set. For Alarm Router the supportability position is the paging provider's older application interface is still in use. Alarm Router pairs a Python 3.11 implementation with cloud, containers plus a scheduled dispatcher, and its data lives in PostgreSQL 14 for routes and delivery receipts. Automated test coverage on Alarm Router stands at 51 per cent line coverage. Alarm Router has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Automated test coverage on Identity and Authorization stands at 84 per cent line coverage. Identity and Authorization commits that token issuance inside 200 ms at p95 and no unauthenticated write path, and the objective is measured continuously rather than sampled. Interface discipline around Identity and Authorization is adequate; the schema is versioned and change is gated on a compatibility check. Identity and Authorization was assessed against the objective and the defect record, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. The Identity and Authorization codebase is Go 1.22 throughout, deployed onto cloud, containers on a managed Kubernetes cluster, and its durable record is PostgreSQL 14 with a hardware-backed key store. Identity and Authorization was tested rather than accepted, and the test is described in the working papers behind TDR-2025-34.

Placement of Identity and Authorization is cloud, global control plane with regional caches, which the reviewer tested by tracing a request end to end. The customer directory federations and the secret store feed Identity and

Authorization; a failure in Identity and Authorization is felt by every cloud service and the remote operations console. Of the 6 open items against Identity and Authorization, the reviewer treats one as material: service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. Identity and Authorization releases weekly, with a security review on every change. Identity and Authorization is owned by the Security Engineering team, with Sonia Krall named as the engineering owner of record. Identity and Authorization is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

For Traffic Manager the reviewer records that no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. The objective recorded against Traffic Manager is that corridor deadlock detected and broken inside two seconds, and it is reported weekly to the engineering leadership. Release cadence for Traffic Manager is weekly, paired with the allocator release, and the release records bear that out. Traffic Manager sits cloud, regional, with a degraded on-robot yield behaviour if the link drops, and no alternative placement is currently supported. Traffic Manager was assessed on the reservation protocol specification and a replay of three deadlock events, and nothing beyond that was available in the data room. Traffic Manager was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Implementation of Traffic Manager rests on Go 1.22; the process runs on cloud, containers on a managed Kubernetes cluster and persists to Redis 7 for corridor reservations with a PostgreSQL 14 audit trail. Traffic Manager depends on the fleet allocator and the site map service and is depended on by the motion planner and the remote operations console. Against Traffic Manager the backlog runs to 12 items, and it includes a known behaviour whereby a corridor reservation leaks when a robot drops off the network mid-traverse. One coupling concern applies to Traffic Manager: Redis 7 for corridor reservations with a PostgreSQL 14 audit trail is shared with a second service, so a schema change is not a local decision. On end-of-life exposure, Traffic Manager shows Redis 6 client library still vendored in one code path. Traffic Manager sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Fleet Health Monitor ships fortnightly; the reviewer sampled the last twelve releases. Fleet Health Monitor does not share a schema with any consumer; every read by the alarm router and the field service workflow goes through the published interface. Fleet Health Monitor sits between the telemetry ingest path and the robot shadow store and the alarm router and the field service workflow, and that position is what makes its availability objective load-bearing. The assessment of Fleet Health Monitor is not unqualified: the design is sound and the test evidence is thinner than the design deserves. 17 defects remain open against Fleet Health Monitor; the one the reviewer pressed on is that alert grouping collapses distinct faults at the same site into one case, hiding the second fault. Fleet Health Monitor is treated as material to the investment case rather than as a routine engineering item.

Supportability of Fleet Health Monitor turns on one fact: none identified. The Field Reliability team carries Fleet Health Monitor on its service inventory, and Wanda Njoku answered for it in interview. Fleet Health Monitor is deployed cloud, regional. Underlying Fleet Health Monitor in this report are the alert rule inventory, three months of case records and a rule coverage review. Test coverage on Fleet Health Monitor is 57 per cent line coverage and the measurement basis was confirmed with Wanda Njoku. Fleet Health Monitor has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2025-34)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	70	0	0.0%	79 min	0
2025-02	56	0	0.0%	68 min	1
2025-03	38	1	2.6%	85 min	1
2025-04	68	1	1.4%	69 min	3
2025-05	69	6	8.6%	45 min	0
2025-06	49	3	6.1%	87 min	1
2025-07	58	3	5.1%	71 min	1
2025-08	48	3	6.2%	30 min	1
2025-09	76	1	1.3%	21 min	3
2025-10	60	4	6.6%	85 min	2
2025-11	59	0	0.0%	30 min	1
2025-12	68	5	7.3%	22 min	2
2026-01	67	6	8.9%	25 min	0

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2026-02	60	0	0.0%	30 min	0
2026-03	42	0	0.0%	51 min	0
2026-04	54	5	9.2%	61 min	1
2026-05	39	2	5.1%	40 min	2
2026-06	67	1	1.4%	45 min	2
2026-07	78	7	8.9%	19 min	3
2026-08	81	4	4.9%	58 min	3

6. Where the incidents landed: sites and customers

Site ST-12 is Newnan, Georgia: Ironwood Grocery Group, 66 robots, live from 2025-07. Newnan, Georgia reports 33% headroom against its integration with vendor B. Newnan, Georgia was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Site ST-17 is Lockbourne, Ohio: Granite Peak Distribution, 19 robots, live from 2026-04. Vendor A is the warehouse system at Lockbourne, Ohio, where headroom is 68%. Lockbourne, Ohio was excluded from an earlier draft of TDR-2025-34 and reinstated after the reviewer obtained the supporting record.

41% of headroom remains at Shepherdsville, Kentucky, whose connector targets vendor C. The Shepherdsville, Kentucky deployment serves Fathom Mobility with 47 robots, live since 2025-03. Shepherdsville, Kentucky is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The Ontario, California site integrates with vendor C and shows 26% headroom. The reason Ontario, California appears here is that it is the site whose shift pattern change caused the April 2025 charge incident. Ontario, California is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

Site ST-02 is Meridian, Idaho: Asterion Logistics, 96 robots, live from 2022-09. On Meridian, Idaho the record shows that it is the longest-running customer deployment and the source of the most incident history. Meridian, Idaho is one of the entries where TDR-2025-34 relies on the company's own measurement rather than an independent one.

Capacity headroom at Fremont, California is not measured, and the warehouse system is test harness. For Fremont, California the material fact is that it is the reference site, and the only one with a laboratory alongside. Fremont, California would be a candidate for the first post-close engineering review under any reasonable plan.

Grand Prairie, Texas is the second Asterion site, commissioned on the same template, and the reviewer weighted the sample accordingly. Grand Prairie, Texas runs 78 robots for Asterion Logistics and opened in 2023-04. Grand Prairie, Texas is treated as material to the investment case rather than as a routine engineering item.

Bethlehem, Pennsylvania came online in 2025-11 and now carries 37 robots for Kestrel Medical Supply. 51% of headroom remains at Bethlehem, Pennsylvania, whose connector targets vendor A. Bethlehem, Pennsylvania carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

Edwardsville, Illinois is notable in this review because it is the newest Crestline site, still ramping. 24 robots are deployed at Edwardsville, Illinois, which went live in 2026-03. Edwardsville, Illinois carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

What distinguishes Lehigh Valley, Pennsylvania is that it is commissioned during the connector rework. Remaining capacity at Lehigh Valley, Pennsylvania is 38%, measured against the site's peak order profile. Lehigh Valley, Pennsylvania is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2025-34)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
Fremont, California	internal engineering site	34	2021-06	not measured	test harness	the reference site, and the only one with a laboratory alongside
Meridian, Idaho	Asterion Logistics	96	2022-09	22%	vendor A	the longest-running customer deployment and the source of the most incident

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
						history
Grand Prairie, Texas	Asterion Logistics	78	2023-04	31%	vendor A	the second Asterion site, commissioned on the same template
Wilmington, Ohio	Crestline Fulfillment	112	2023-08	14%	vendor B	the largest deployment and the closest to the allocator scaling limit
Lehigh Valley, Pennsylvania	Crestline Fulfillment	64	2024-02	38%	vendor B	commissioned during the connector rework
Ontario, California	Evergreen Warehousing	88	2024-05	26%	vendor C	the site whose shift pattern change caused the April 2025 charge incident
Groveport, Ohio	Granite Peak Distribution	52	2024-09	44%	vendor A	opportunity charging only, which is where the pack estimator drift appeared
Kenosha, Wisconsin	Dunbar Industrial	41	2024-11	49%	vendor D	the second opportunity-charging site
Tracy, California	Harborline Systems	58	2025-01	36%	vendor B	the site whose cleaning schedule now overlaps the peak shift
Shepherdsville, Kentucky	Fathom Mobility	47	2025-03	41%	vendor C	manipulator-equipped robots are concentrated here
Reno, Nevada	Blue Mesa Components	29	2025-05	57%	vendor D	the smallest customer deployment
Newnan, Georgia	Ironwood Grocery Group	66	2025-07	33%	vendor B	narrow-aisle geometry, and the source of the August 2025 mutual-yield stall
Whitestown, Indiana	Juniper Bay Apparel	44	2025-09	45%	vendor C	the site with the standing map publication workaround
Bethlehem, Pennsylvania	Kestrel Medical Supply	37	2025-11	51%	vendor A	sparse survey features, which is why relocalisation is slow here
Chino, California	Larkspur Home	33	2026-01	54%	vendor D	the go-live delayed by the forked provisioning template
Edwardsville, Illinois	Crestline Fulfillment	24	2026-03	63%	vendor B	the newest Crestline site, still ramping
Lockbourne, Ohio	Granite Peak Distribution	19	2026-04	68%	vendor A	commissioned after the template baseline check landed
Mesquite, Texas	Evergreen Warehousing	16	2026-06	71%	vendor C	in acceptance testing at the review date

7. Security findings bearing on availability

Reading SEC-2026-09, the defect is that fleet health monitor query interface allowed unbounded result sets; the discovery date is 2026-04-18. Residual exposure on SEC-2026-09 against Fleet Health Monitor reads low. Evidence behind SEC-2026-09 against Fleet Health Monitor is that service restart induced by the tester's query, which the reviewer treats as moderate. On Low-severity finding SEC-2026-09 the conclusion is that the severity is right and the ownership is unclear, since two teams each believe the other holds it. finding SEC-2026-09 is treated as material to the investment case rather than as a routine engineering item.

The finding at assessment finding SEC-2025-05, dated 2025-09-19, is that telemetry ingest accepted a robot identity without checking device attestation. Strong evidence supports SEC-2025-05, namely that the tester submitted synthetic telemetry from an unenrolled host. What remains after High-severity finding SEC-2025-05 is remediated is low; attestation is now required on the ingest path. In assessment finding SEC-2025-05 the exploitation path means a forged agent could inject telemetry attributed to a real robot. Telemetry Ingest finding SEC-2025-05 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

High-severity finding SEC-2026-01 is rated High with a numeric band of 7.1 on the tester's scoring scale. High-severity finding SEC-2026-01 records that emergency release path permits a single approver, discovered on 2026-04-14. Evidence behind finding SEC-2026-01 is that the tester used the path in a staging tenant and the audit trail recorded no justification, which the reviewer treats as limited, resting on a configuration read rather than on a demonstrated exploitation. In High-severity finding SEC-2026-01 the exploitation path means one insider can move an unreviewed artefact to production. SEC-2026-01 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

For SEC-2026-19 the reviewer notes that the finding was closed within the committed window and the closure evidence was inspected. Medium is the rating SEC-2026-19 against Localization Engine carries, at 5.1 on the assessor's own scale. Impact of Localization Engine finding SEC-2026-19 is not theoretical; a stolen compute module discloses a customer's warehouse layout. After the fix, Localization Engine finding SEC-2026-19 carries residual risk described as medium; disk encryption is planned for the next compute revision. Localization Engine finding SEC-2026-19 was reviewed against the position twelve months earlier, and nothing measurable has changed.

The practical consequence in SEC-2026-03 against Config Distribution is that a local compromise on a robot becomes a persistent configuration compromise. The evidence for SEC-2026-03 against Config Distribution is moderate: the tester applied an unsigned bundle on a laboratory robot. High-severity finding SEC-2026-03 records that config bundle signature verification could be disabled by a local flag, discovered on 2026-04-15. The remediation ledger shows Config Distribution finding SEC-2026-03 as remediated, 2026-06-12. SEC-2026-03 was checked against the corresponding entry in the company's own register, and the register was current.

For Video Snippet Service finding SEC-2025-06 the reviewer notes that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. The evidence for SEC-2025-06 against Video Snippet Service is moderate, because the demonstration was performed in a laboratory tenant rather than in production: enumeration confirmed; no clip content was retrievable. On 2025-09-19 the tester recorded Medium-severity finding SEC-2025-06: video snippet index allowed cross-tenant enumeration by identifier. What SEC-2025-06 against Video Snippet Service enables is this: a customer operator could enumerate the existence of clips in another tenant. SEC-2025-06 against Video Snippet Service is the clearest instance in TDR-2025-34 of engineering practice being better than the documentation suggests.

Committed 2026-09-30 is the closure position on SEC-2026-05 against Alarm Router, which the ledger records as open. Rating for finding SEC-2026-05 is Medium, band 5.2; the reviewer re-read the scoring rationale and accepted it. Medium-severity finding SEC-2026-05 produces this judgement: the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Impact of SEC-2026-05 against Alarm Router is not theoretical; one leaked credential affects paging for every region. Medium-severity finding SEC-2026-05 has no single accountable owner, and that is the finding the reviewer would press hardest.

SEC-2025-11 against Remote Operations Console leaves residual risk that the reviewer assesses as low. In Low-severity finding SEC-2025-11 the exploitation path means a cross-site request could reuse an active operator session.

Rating for Low-severity finding SEC-2025-11 is Low, band 3.1; the reviewer re-read the scoring rationale and accepted it. Finding SEC-2025-11 sits against Remote Operations Console and records that console session cookie lacked a strict same-site attribute. finding SEC-2025-11 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Reading finding SEC-2026-04, the defect is that wMS connector logged customer order payloads at debug level; the discovery date is 2026-04-16. Assessment finding SEC-2026-04 matters because customer commercial data is retained in operational logs. Moderate evidence supports finding SEC-2026-04, namely that payloads recovered from a staging log store. Tracking finding SEC-2026-04 to closure gives a status of remediated and a date of 2026-05-20. SEC-2026-04 against WMS Connector has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Low-severity finding SEC-2026-12 records that edge gateway agent retained mutual authentication keys after decommissioning, discovered on 2026-04-20. SEC-2026-12 against Edge Gateway Agent leaves residual risk that the reviewer assesses as medium; a wipe procedure exists but was not applied retroactively. SEC-2026-12 stands partially remediated; pending is what the record gives for closure. Low-severity finding SEC-2026-12 matters because a disposed compute module carries a usable fleet credential. SEC-2026-12 against Edge Gateway Agent carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

Not planned is the closure position on SEC-2026-20 against Payload Scale Interface, which the ledger records as accepted. The reviewer's position on SEC-2026-20 against Payload Scale Interface is that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. In SEC-2026-20 against Payload Scale Interface the exploitation path means a calibration change cannot be reconstructed during an investigation. Reading Payload Scale Interface finding SEC-2026-20, the defect is that payload scale calibration record can be edited without an audit entry; the discovery date is 2026-04-24. Payload Scale Interface finding SEC-2026-20 is the clearest instance in TDR-2025-34 of engineering practice being better than the documentation suggests.

Residual risk after remediation of Charge Scheduler finding SEC-2025-10 is low. Assessment finding SEC-2025-10 is rated Low with a numeric band of 3.7 on the tester's scoring scale. As at the review date SEC-2025-10 against Charge Scheduler was remediated and the closure entry reads 2025-10-09. On 2025-09-24 the tester recorded SEC-2025-10: charge scheduler exposed an unauthenticated internal health endpoint with site names. SEC-2025-10 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

8. Reliability debt and what it would cost to clear

Reading IP and support item TD-02 against the funded plan, the reviewer treats this as both an IP and a supportability item. Exposure carried by Localization Engine item TD-02 is that it strong-copyleft exposure in shipped robot software plus an unsupported dependency. TD-02 on Localization Engine is funded by reallocating effort from feature work; the register status and the funding position are not the same thing. TD-02 on Localization Engine was one of two items where TDR-2025-34 could not obtain the underlying data within the engagement.

Reading the programme plan, TD-23 on Charge Scheduler is behind a supplier release outside the company's control. Item Charge Scheduler item TD-23 in the debt register is to split the shared database instance used by the allocator and the charge scheduler, and it bears on Charge Scheduler. Whether reliability item TD-23 happens depends on funding, and it is funded from the second quarter onward. register item TD-23 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Item programme item TD-10 in the debt register is to migrate the deprecated warehouse transformation function, and it bears on Analytics Warehouse Loader. Reading TD-10 against the funded plan, the warehouse vendor has published a withdrawal date. Status on Analytics Warehouse Loader item TD-10 is committed; the target is 2026-12-31. support item TD-10 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

TD-21 on Battery Management Service is behind a platform change that has its own target date, which is why the target date is later than the effort estimate implies. The business consequence recorded against Battery Management Service item TD-21 is that it cell imbalance can hide until derating. In progress is the status of register item TD-21, targeted at 2026-09-30. Battery Management Service item TD-21 is the entry that changed most between the first and second document requests behind TDR-2025-34.

TD-19 is behind a platform change that has its own target date, which is why the target date is later than the effort estimate implies. Measure closing rate for the separation monitor is entered at TD-19 as an safety item. Register item TD-19 carries a 2026-08-31 target and a status of in progress. safety item TD-19 is the clearest instance in TDR-2025-34 of a control that exists on paper and is weaker in practice.

Under safety, entry safety item TD-20 is to validate and ship the wheel slip detector. Progress on programme item TD-20 is recorded as in progress against a 2026-09-15 date. The estimate for Wheel Odometry Filter item TD-20 is 1.5 engineer-quarters and $162,000. safety item TD-20 was checked against the company's own register, which had not been updated since the change landed.

The register shows Simulation Harness item TD-06 as blocked, due 2026-11-30. The business consequence recorded against TD-06 on Simulation Harness is that it a nightly regression can reach a release candidate. TD-06 on Simulation Harness is not straightforward: blocked on the physics core determinism work. TD-06 is the clearest instance in TDR-2025-34 of engineering practice being better than the documentation suggests.

Programme item TD-18 is sequenced behind a platform change that has its own target date. TD-18 on Safety Controller carries a 2027-03-31 target and a status of backlog. TD-18 matters because it investigation evidence is lost between weekly extractions. Safety Controller item TD-18 is treated as material to the investment case rather than as a routine engineering item.

If TD-08 on Remote Operations Console is not done, the exposure is that it no upstream security updates for a customer-facing component. TD-08 on Remote Operations Console is the item to replace the archived charting library in the operator console, classified under security supportability. TD-08 on Remote Operations Console is sized at 1 engineer-quarter, or $132,000 at the company's loaded rate. programme item TD-08 would move from a concern to a verified defect if one further measurement were supplied.

Behind a platform change that has its own target date is where programme item TD-11 sits in the programme. Re-converge the four forked site provisioning templates is entered at programme item TD-11 as an configuration item. Status on register item TD-11 is in progress; the target is 2026-09-30. programme item TD-11 has no single accountable owner, and that is the finding the reviewer would press hardest.

9. Incidents with a safety dimension

HZ-04 applies to robots fitted with the weighing deck. Remote-exposure hazard HZ-04 is the entry for the case that gripper closes on a hand during a manual intervention. The verification method for Serious injury hazard HZ-04 is force bench test after every calibration, evidenced by a calibration record. Serious injury hazard HZ-04 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Severity at HZ-11 is Fire, with the exposure rated Improbable and controllability Low. HZ-11 is addressed by cell temperature limits, pack-level current derating and a dock thermal cutout, and the reviewer traced that control into the design record. What the register does not say about Improbable-exposure hazard HZ-11, and the reviewer does, is that state-of-health drift on opportunity-charged packs is an open modelling item. Fire hazard HZ-11 is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

Register entry HZ-03 depends for its mitigation on payload mass and centre-of-gravity limits enforced by the scale interface. Hazard HZ-03 in the register covers the case where load falls from an elevated deck during transport. Hazard HZ-03 carries residual risk of Tolerable where the weighing deck is fitted with a status of partially verified. register entry HZ-03 is reported at the evidence strength available and has not been promoted to fit the narrative.

No direct harm hazard HZ-17 carries residual risk of Tolerable, with an evidence-loss caveat with a status of open. Hazard HZ-17 in the register is not clean: extraction runs weekly, so a busy robot can still lose events between extractions. On the register's own scale, Frequent-exposure hazard HZ-17 is No direct harm severity, Frequent exposure, Low controllability. Frequent-exposure hazard HZ-17 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Produced two near-miss reports from site staff is the incident history behind register entry HZ-14. One qualification on Occasional-exposure hazard HZ-14 the reviewer would not remove is that the roster invalidation fix landed in July

2025. Where HZ-14 bites is the sites that have moved to opportunity charging. Occasional-exposure hazard HZ-14 is included because a reader of TDR-2025-34 should be able to disagree with the inference without repeating the work.

Status verified and residual risk Broadly acceptable are what the register gives for hazard HZ-21. Remote-exposure hazard HZ-21 is relevant to every robot variant currently in production rather than to the whole fleet. HZ-21 is addressed by push detection that engages the brake and requires a deliberate release, and the reviewer traced that control into the design record. hazard HZ-21 was one of two items where TDR-2025-34 could not obtain the underlying data within the engagement.

The residual entry for Serious injury hazard HZ-19 reads Tolerable, verified. Protection at Remote-exposure hazard HZ-19 comes from dock occupancy sensing plus a low approach speed inside the dock envelope. Hazard HZ-19 applies to the two pallet-mover variants only. Serious injury hazard HZ-19 is the clearest instance in TDR-2025-34 of engineering practice being better than the documentation suggests.

Occasional-exposure hazard HZ-13 is mitigated by detection of unmapped obstacles with a stop, plus a site procedure for barriers. Field history for hazard HZ-13 is that it has produced no event, though the control has activated repeatedly. Hazard HZ-13 remains Tolerable where the barrier meets the visibility guidance after the control is applied, and the register marks it partially verified. hazard HZ-13 carries a follow-up request that was satisfied after the draft of TDR-2025-34 was circulated.

Serious injury hazard HZ-02 is mitigated by aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. Hazard HZ-02 in the register is relevant to the sites that have moved to opportunity charging rather than to the whole fleet. Severity at hazard HZ-02 is Serious injury, with the exposure rated Occasional and controllability Low. hazard HZ-02 in the register was inspected in the repository rather than described from the ticket, which is why the account is specific.

Risk parameters for hazard HZ-15 in the register are Serious injury severity, Remote exposure and Medium controllability. Verification of Remote-exposure hazard HZ-15 was by site commissioning check plus a fire drill observation at two sites. Register entry HZ-15 has produced no event, though the control has activated repeatedly, and the register was not updated to reflect it. register entry HZ-15 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

10. Unresolved matters and management's declinatures

The reviewer has resisted the temptation to tidy the report at TDR-2025-34: the open items are open because the evidence is not there yet. Report TDR-2025-34 closes with 9 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Deal implications in the report at TDR-2025-34 are modest in aggregate and specific in detail: 3 items for the closing checklist, 6 for the plan. The costed remediation implied by the present report is $1,530,000 over the coming four quarters, which is within the ordinary engineering budget. Readers comparing report TDR-2025-34 with an earlier technical report should note that the service inventory was renamed in between.

R. Ostrowski asked in every declined case who would revisit the decision and when, and TDR-2025-34 records the answer even where there was none. Two positions in this report are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. Method in this report was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. R. Ostrowski was granted read access to the code, the pipeline and the incident tracker for the purposes of TDR-2025-34; production access was not requested. Cross-references in this report point to the evidence index rather than to a summary of the evidence.

The present report recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Reporting under the report at TDR-2025-34 should distinguish a target that was met from a target that was lowered, which the current pack does not. 6 sites contributed evidence to this report, and one further site was in acceptance testing and was excluded. The population for report TDR-2025-34 was the service inventory as at 2026-01-09; 20 entries were examined and the rest were reviewed only at inventory level. R. Ostrowski notes that the company answered every document request, though two answers arrived after the draft.

In report TDR-2025-34 the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. This report grades evidence as strong, moderate or limited, and a limited grade is not

upgraded because the consequence would be serious. The open items in report TDR-2025-34 are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. The report at TDR-2025-34 preserves 9 unresolved questions rather than resolving them by assumption. The company reviewed a draft of the present report for factual accuracy and its corrections are reflected without altering any conclusion.

11. Conclusions on reliability and post-close monitoring

Calder Rowe Engineering Advisory concludes, on the material examined, that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The judgement report TDR-2025-34 reaches is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Three of the open items in report TDR-2025-34 would close on a single document that the company has agreed to produce. 9 matters remain open at the date of this report, and none of them is closed by the passage of time alone. Calder Rowe Engineering Advisory has costed remediation at the company's own loaded engineering rate rather than at a market rate.

For the transaction, this report identifies 3 matters the reviewer would raise before closing and 6 for the post-close engineering plan. TDR-2025-34 separates matters that bear on the closing conditions from matters that belong in the first two quarters after close. R. Ostrowski's conclusion in the report at TDR-2025-34 is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The report at TDR-2025-34 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The reliability record, incident history and post-incident follow-through of the fleet platform is a wide field, and TDR-2025-34 is explicit about which parts of it were reached.

R. Ostrowski would add one qualitative item to the pack proposed in this report: the count of commitments closed without verifying evidence. R. Ostrowski notes in the present report that the company's own reporting is adequate for three of the four recommended indicators. Taking the sections above together, there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. On the evidence available by 2026-01-09, this report concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The present report contains 9 open items, and a reader should assume that number will move rather than settle.